

6G Weekly Survey — May 25 - May 31, 2026

18 items across 4 topics, ranked by importance.

AI-native air interface

TONIC: Token-Centric Semantic Communication for Task-Oriented Wireless Systems

Sige Liu, Kezhi Wang · arXiv cs.IT · 2026-05-20

★★★★■ Addresses the fundamental mismatch between bit-centric wireless design and token-based AI inference with a principled, modular architecture that could inform task-oriented communication standards as foundation models become wireless endpoints.

TONIC introduces a token-centric semantic communication framework where the transmitter allocates unequal error protection based on token-level task relevance, and the receiver uses confidence-gating to mask unreliable tokens before Transformer-based completion restores them for inference. Unlike fully end-to-end learned approaches, TONIC combines semantic-aware protection with interpretable modular components including utility-aware Bayes-risk gating. Experiments on image classification demonstrate consistent gains over DeepJSCC and separation-based schemes across AWGN, Rayleigh, and Rician channels.

Contribution: First framework to explicitly bridge token-level representation in foundation models with wireless communication design through utility-aware unequal error protection and confidence-gated token completion, moving beyond bit-fidelity and pixel-domain semantic communication.

Relevance to your work: Directly relevant to your semantic and task-oriented communication research, offering a concrete bridge between AI-native air interfaces and downstream model requirements through interpretable token-level protection rather than black-box end-to-end learning.

Resilience Characterization of AI-Native Wireless Receivers via Persistent Homology

Christo Kurisummootil Thomas, Emilio Calvanese Strinati · arXiv cs.IT · 2026-05-21

★★★★■ Addresses a critical gap in deploying AI-native receivers—lack of real-time resilience metrics—with mathematically principled methods and demonstrates practical early-warning advantages over standard gradient-based monitoring.

The paper introduces a Topological Resilience Index (TRI) based on persistent homology to quantify the real-time resilience of neural network wireless receivers during online adaptation to non-stationary channels. TRI combines three dimensions—validation-loss topology, channel impulse response distribution drift, and channel manifold geometry—to provide early warning of performance degradation. Simulations on OFDM receivers across ITU-R environment transitions show TRI provides >1 OFDM symbol advance warning over conventional metrics and enables 80% BER reduction through guided re-adaptation.

Contribution: First application of persistent homology and topological data analysis to characterize neural receiver resilience in real-time, with provable bounds on stability and early-warning capability for channel distribution shifts.

Relevance to your work: Directly relevant to AI-native air interface development: provides a rigorous framework for monitoring and adapting learned PHY layers under non-stationary channels, essential for 6G deployment where neural receivers must handle diverse and dynamic propagation environments.

6G Physical Layer

ComHymba: Low-Complexity Domain-Informed Foundation Model for Wireless Communications

Bowen Yang, Wei Chen, Jiaming Cheng, Bo Ai · arXiv eess.SP · 2026-05-22

★★★★■ Addresses critical computational bottleneck of Transformer-based wireless models with a principled hybrid architecture and demonstrates practical gains on a representative task suite relevant to 6G AI-native PHY.

ComHymba is a wireless foundation model using asymmetric masked autoencoders for self-supervised pre-training on CSI, replacing quadratic-complexity Transformers with Hymba blocks that combine windowed self-attention and state-space models for linear-time scaling. It incorporates domain knowledge through 3D spatio-temporal-frequency patchification, physics-aware masking strategies that mimic fading patterns, and a decoupled amplitude-phase loss function aligned with channel statistics. Evaluation across eight downstream tasks (channel reconstruction, sensing, beam management) demonstrates accuracy improvements over task-specific baselines with 3.3× faster inference than Transformer equivalents.

Contribution: First wireless foundation model to integrate state-space models with domain-informed masking and propagation priors, achieving linear complexity in CSI sequence length while maintaining strong performance across diverse physical-layer tasks.

Relevance to your work: Directly applicable to your AI-native air interface research as a scalable pre-training backbone for learned PHY components including CSI compression, beam management, and ISAC sensing tasks.

AFDM as a Software Upgrade of OFDM: One Firmware Patch, a New Frontier

Hyeon Seok Rou, Giuseppe Thadeu Freitas de Abreu · arXiv eess.SP · 2026-05-21

★★★★■ Strong practical result from credible authors showing deployability barrier for AFDM is lower than believed, though real-world validation and industry adoption remain to be demonstrated.

The authors demonstrate that affine frequency division multiplexing (AFDM) can be implemented as a firmware patch over existing OFDM infrastructure with marginal cost, enabling robustness to doubly dispersive channels, inherent ISAC compatibility, and physical-layer security. They also prove AFDM achieves full uncoded diversity over static LTI channels, revealing previously unexploited multicarrier capabilities. The work positions AFDM as a practical near-term upgrade path for deployed systems rather than requiring new hardware.

Contribution: Proof that AFDM requires only software modification of OFDM systems rather than hardware replacement, making advanced 6G waveform features deployable on existing infrastructure.

Relevance to your work: Directly relevant to your 6G waveform research (AFDM) and ISAC work, as this addresses the critical deployment feasibility question and confirms AFDM's inherent sensing-communication compatibility.

A Measurement-Based Parameterization of Physics Reflection Models for Terahertz Communication

Taihao Zhang, Chenzhou Lin, Cunhua Pan, Hong Ren, et al. · arXiv eess.SP · 2026-05-22

★★★★■ Addresses a critical gap in THz channel modeling with direct measurements at frequencies being considered for 6G, providing validated physics-based reflection models essential for accurate ray-tracing and site-specific propagation prediction in sub-THz deployments.

The authors build a 300-400 GHz measurement platform to characterize reflection coefficients of various materials, propose a single-layer interference model with extended-parameterized Lorentz/Drude (SLI-EPLD) formulation using sub-band modeling and parameterized mapping for frequency-dependent variation, and introduce a weighted sub-band fitting algorithm (WF-TREND) for parameter extraction. Validation against measurements shows improved accuracy over existing reflection models for multiple materials. This provides a physics-based reflection model for high-THz channel modeling in beyond-5G systems.

Contribution: First measurement-backed parameterization of reflection coefficients in the 300-400 GHz band using a physics-informed SLI-EPLD model with stable sub-band parameter extraction, directly supporting channel modeling at frequencies relevant to 6G sub-THz bands.

Relevance to your work: Directly supports your work on 6G waveforms and sub-THz physical layer design by providing measurement-validated material reflection models required for accurate channel simulation,

ray-tracing environments, and potential integration with GAN-based generative channel models at these frequencies.

DAFT-s-AFDM Enabled ISAC Systems: Ambiguity Function Analysis and Waveform Design

Shiqi Cui, Tianqi Mao, Fan Zhang, Zeping Sui, et al. · arXiv eess.SP · 2026-05-19

★★★★■ Strong technical contribution addressing two critical 6G topics (AFDM waveforms and ISAC) with rigorous theoretical analysis and practical optimization framework from a credible research group.

This paper analyzes DAFT-spread-AFDM waveforms for ISAC systems, deriving closed-form expressions for the ambiguity function magnitude expectation and revealing how waveform parameters affect sensing-communication tradeoffs. The authors propose a probabilistic constellation shaping (PCS) framework that jointly optimizes communication throughput and sensing AF characteristics via multi-objective optimization with Pareto-optimal solutions. Simulations demonstrate PCS-enhanced DAFT-s-AFDM achieves superior and controllable dual-functional performance versus classical approaches.

Contribution: First closed-form ambiguity function analysis for DAFT-s-AFDM enabling parameter-driven ISAC waveform design, coupled with a novel PCS framework for joint sensing-communication optimization.

Relevance to your work: Directly relevant to your ISAC and 6G waveform interests, particularly the analytical characterization of AFDM sensing performance and multi-objective optimization methods for dual-functional systems.

SPA-MAE: A Physics-Guided CSI Foundation Model for Wireless Physical Layer

Chen Chen, Weijie Jin, Hengtao He, Xiaoheng Sun, et al. · arXiv cs.IT · 2026-05-19

★★★★■ Strong contribution addressing a critical gap in wireless foundation models by grounding pretraining in channel physics rather than generic vision model adaptation, demonstrating measurable gains in practical low-resource scenarios relevant to 6G deployment.

SPA-MAE is a wireless foundation model for CSI that incorporates physics-guided pretraining through two complementary branches: parameter-aware guidance that aligns encoder features with explicit multipath parameters, and structure-aware guidance that captures sparse transformed-domain CSI structure via 2D FFT. The pretrained MAE encoder generalizes across four downstream physical layer tasks, outperforming existing CSI foundation models with fewer parameters, particularly in low-SNR and data-limited regimes.

Contribution: First CSI foundation model to systematically inject dual physics priors (multipath parameters and sparse transform structure) into MAE pretraining rather than relying solely on reconstruction objectives or naively adapting vision models.

Relevance to your work: Directly relevant to AI-native air interface research: provides a principled approach to building foundation models for learned PHY that could underpin neural receivers, CSI compression, and channel estimation tasks in your portfolio.

ISAC (sensing + communication)

Diffusion Fluid Antenna Systems for Resilient ISAC

Noor Waqar, Kai-Kit Wong, Chan-Byoung Chae, Ross Murch · arXiv eess.SP · 2026-05-22

★★★★■ Novel integration of diffusion models with FAS for ISAC addresses an underexplored object-side perspective with strong experimental results, though practical deployment complexity and scalability remain open questions.

This paper introduces diffusion FAS (fluid antenna systems) for ISAC, using a conditional DDPM to reconstruct spatial channel structure from sparse port measurements and enable dynamic port selection. Rather than optimizing waveform or power allocation, the approach exploits reconfigurable spatial DoF to either suppress a target's radar cross-section (generative spatial stealth) or null interference from adjacent objects (target isolation). Demonstrations show sensing visibility reduction by up to two orders of magnitude.

Contribution: First application of generative diffusion models to fluid antenna port selection for ISAC, enabling object-side control of sensing signatures through learned spatial agility rather than traditional BS-centric waveform design.

Relevance to your work: Directly relevant: combines generative AI (diffusion models) with ISAC and introduces a new spatial DoF that complements your work on AI-native air interfaces and next-generation sensing-communication co-design.

FMCW-Based Integrated Sensing and Communication System: Design, Implementation, and Experimental Measurements

Murat Temiz, Colin Horne, Matthew A. Ritchie, Christos Masouros · arXiv eess.SP · 2026-05-22

★★★★■ Strong practical contribution with hardware validation addressing the critical radar-first ISAC paradigm for vehicular networks, directly relevant to 3GPP Release 19+ sensing discussions.

This paper presents a radar-centric ISAC system where FMCW chirps carry data via two-layer modulation (phase modulation and index modulation) while preserving radar sensing as primary function. A novel signal processing technique mitigates modulation impacts on sensing accuracy, and a custom receiver demodulates data from chirps. Hardware proof-of-concept at 2.4/24 GHz demonstrates 25-50 Mbps throughput with experimental loopback measurements validating the architecture.

Contribution: The dual-layer PM+IM modulation scheme on FMCW chirps with dedicated mitigation algorithms enables practical radar-first ISAC with measured hardware validation, unlike most simulation-only work.

Relevance to your work: Directly relevant: FMCW-based ISAC with experimental validation addresses fundamental waveform design questions for joint sensing/communication that inform your work on AI-native air interfaces and integrated sensing paradigms.

Reliable UAV Detection with ISAC

Stephan Saur, Mark Doll, Artjom Grudnitsky, Silvio Mandelli, et al. · arXiv eess.SP · 2026-05-22

★★★★■ Strong practical validation of ISAC on commercial 5G systems with real performance data, directly relevant to 3GPP Release 19 ISAC standardization and demonstrating path to deployment without custom hardware.

This paper experimentally validates UAV detection using unmodified commercial 5G hardware as a monostatic OFDM radar, comparing real-world performance against theoretical link budget and hardware impairment models. The authors demonstrate reliable detection with sub-meter accuracy at distances exceeding 500 meters in clutter-rich environments. The work bridges theoretical ISAC models with practical deployments on commercial 5G infrastructure.

Contribution: First experimental demonstration that commodity 5G hardware can reliably detect small UAVs at 500+ meter ranges with sub-meter accuracy despite hardware impairments and clutter, validating ISAC feasibility without specialized radar equipment.

Relevance to your work: Directly aligns with your ISAC research interests by providing empirical benchmarks for sensing performance on commercial OFDM systems, informing design tradeoffs between communication and sensing functions in AI-native air interfaces.

Communication Security and Sensing Privacy in FMCW-Based ISAC Through Signal Modulation

Murat Temiz, Christos Masouros · arXiv eess.SP · 2026-05-22

★★★★■ Addresses a critical but under-explored problem in ISAC—securing sensing information from eavesdroppers—with a practical FMCW implementation and complete transceiver architecture from credible authors in the field.

The paper proposes a secure ISAC framework using FMCW radar with index modulation for communication security and phase coding to deliberately perturb the ambiguity function, preventing unauthorized sensing eavesdroppers from estimating target velocity and degrading range estimation. The radar-centric design

includes transmitter/receiver architectures for modulation/demodulation and legitimate sensing operations. Simulations demonstrate high data throughput while enhancing both communication security and sensing privacy against passive adversaries.

Contribution: Novel dual-layer security approach combining index modulation for data protection with deliberate ambiguity function perturbation via phase coding to achieve sensing privacy, specifically preventing velocity estimation by unauthorized receivers.

Relevance to your work: Directly relevant to your ISAC research, particularly the physical-layer security dimension that becomes critical as ISAC moves toward standardization in 6G, where sensing privacy will be a regulatory and commercial concern.

Rate-Splitting--Inspired Bistatic OFDM-ISAC

Bruno F. Costa, Anup Mishra, Israel Leyva-Mayorga, Taufik Abrão, et al. · arXiv eess.SP · 2026-05-20

★★★★■ Addresses a fundamental interference problem in OFDM-ISAC with a novel rate-splitting approach from credible researchers (Popovski group), providing practical optimization framework for dual-functional radar-communication systems relevant to 6G.

The paper tackles bistatic OFDM-ISAC uplink where unequal direct/echo paths and Doppler-induced ICI create challenging interference that standard orthogonal separation or successive interference cancellation cannot handle. The authors propose a rate-splitting framework where communication messages are split into robust and supplementary streams, jointly superposed with sensing signals, and processed by a staged sensing-communication receiver. Convex optimization yields power allocation that maximizes spectral efficiency under sensing-performance constraints, outperforming NOMA baselines in interference management and Doppler robustness.

Contribution: First application of rate-splitting principles to bistatic OFDM-ISAC with tractable per-subcarrier SINR analysis and Fisher-information-based sensing-communication performance tradeoff quantification.

Relevance to your work: Directly addresses ISAC interference mitigation in OFDM, complementing your work on AI-native air interfaces by providing analytical frameworks for joint sensing-communication optimization that could inform learned PHY design or hybrid classical-neural architectures.

Stacked Intelligent Metasurfaces for Resolution-Constrained Near-Field Range Extension in 6G Systems

Yajun Zhao · arXiv cs.IT · 2026-05-19

★★★★■ Strong theoretical contribution addressing a critical bottleneck in near-field ISAC systems with practical design framework, though experimental validation and comparison with alternative beamforming architectures would strengthen commercial relevance.

The paper addresses near-field electromagnetic focusing for 6G ISAC by introducing stacked intelligent metasurfaces (SIMs) to extend the usable near-field distance (UNFD) beyond single-layer limitations. Through unified operator-approximation framework and analytical models, the authors show that multilayer stacking improves wavefront curvature matching and reduces phase error, pushing UNFD closer to the Rayleigh distance while revealing fundamental distance-resolution tradeoffs where axial resolution degrades quadratically. Simulations confirm performance gains saturate due to accumulated losses.

Contribution: Introduces the engineering metric of usable near-field distance (UNFD) distinct from Rayleigh distance, and demonstrates how multilayer SIM architectures systematically extend this range through cascaded wavefront control despite inherent quadratic axial resolution degradation.

Relevance to your work: Directly relevant to ISAC research agenda, particularly for near-field sensing and beam management in massive MIMO/RIS-assisted 6G systems where understanding fundamental range-resolution tradeoffs and multilayer reconfigurable surface architectures is essential for practical deployment.

Movable-Antenna-Enhanced ISAC: Optimal Antenna Trajectory and Beamforming Design

Yifei Wu, Dongfang Xu, Derrick Wing Kwan Ng, Wolfgang Gerstaecker, et al. · arXiv eess.SP · 2026-05-22

★★★★ Solid theoretical contribution to MA-ISAC with rigorous optimization framework, but practical implementation of physically moving antennas at RF timescales remains highly speculative and distant from commercial deployment. This paper proposes a movable antenna (MA) system for ISAC where antennas physically move along continuous trajectories during transmission, creating a synthetic aperture that improves angular resolution for sensing. The authors jointly optimize antenna motion trajectories and transmit beamforming to minimize sensing beam pattern error while maintaining communication QoS constraints, using a branch-and-bound algorithm for globally optimal solutions. Simulations demonstrate substantial gains over static or single-step antenna repositioning schemes.

Contribution: First treatment of continuous antenna trajectory optimization for ISAC that exploits spatio-temporal sampling to synthesize large virtual arrays, moving beyond prior MA work that only considered discrete repositioning.

Relevance to your work: Directly addresses ISAC design trade-offs and introduces a novel spatial degree-of-freedom that could complement beamforming and CSI-based approaches you work on, though the physical feasibility of sub-wavelength antenna motion warrants skepticism.

Energy-Efficient THz Sensing with Hybrid THz/VLC Communication Under Human Blockage Effects

Hanshita Prabhakar, Neetu R. R., Vivek Ashok Bohara · arXiv eess.SP · 2026-05-22

★★★★ Solid work addressing practical THz ISAC deployment challenges (blockage, energy efficiency) with hybrid VLC backup, though limited to indoor scenarios and incremental methodologically.

This paper proposes a hybrid THz/VLC indoor system where THz handles both sensing and high-capacity communication while VLC provides fallback connectivity under blockage. The authors optimize power allocation between THz sensing and communication APs based on monostatic sensing metrics (detection probability, false alarm, sensing capacity), then use MILP to select active VLC APs and minimize network power consumption. Simulations show VLC takes over when human blockages degrade THz links, maintaining energy-efficient coverage.

Contribution: First integrated optimization of THz-based ISAC with VLC fallback, explicitly accounting for human blockage effects on both sensing performance and communication mode selection.

Relevance to your work: Directly relevant to your ISAC interests—demonstrates how sensing requirements shape resource allocation in hybrid communication architectures, with blockage modeling applicable to sub-THz 6G scenarios.

Preisach Attention: A Hysteretic Model of Sequential Memory

Piotr Frydrych · arXiv cs.LG · 2026-05-22

★★★★ Novel mathematical formulation from physics, but lacks wireless domain validation and unclear how hysteretic memory benefits PHY-layer tasks versus standard transformers.

Introduces Preisach Attention Layer (PAL), a sequence modeling architecture based on hysteresis operators that replaces softmax attention with binary relays parameterized by learned thresholds and maintains a stack of local extrema. Proves single-layer PAL is Turing-complete under arbitrary precision and computes historical range statistics in $O(1)$ layers versus $O(\log n)$ for transformers, though lacking random-access retrieval. Achieves $O(n \log n)$ inference cost versus $O(n^2)$ for standard attention, suited for long episodic memory with weak positional dependence.

Contribution: First application of Preisach hysteresis operators to sequence modeling, achieving Turing-completeness in $O(1)$ depth and rate-independent memory through extremum tracking rather than positional encoding.

Relevance to your work: Tangential—while rate-independent memory could theoretically apply to channel state tracking or sensing sequences invariant to sampling rates, the paper presents no wireless applications and the extremum-based attention mechanism does not obviously map to CSI processing, beam management, or ISAC fusion tasks.

UAV-based Energy-Efficient Data Collection in Smart Grids with ISAC QoS Guarantees

Yibin Xie, Jin Zhao, Indrakshi Dey, Nicola Marchetti · arXiv eess.SP · 2026-05-21

★★★★■ Competent application-driven work demonstrating ISAC benefits in a specific vertical (smart grids), but the ISAC modeling appears relatively simple and the core algorithmic contributions are incremental.

This paper addresses UAV-based data collection from power grid monitoring devices using integrated sensing and communication (ISAC) to ensure safe cooperative flight and reliable monitoring. The authors formulate a joint optimization problem for UAV trajectory and scheduling under ISAC quality-of-service constraints, solving it via hybrid deep reinforcement learning and semidefinite relaxation. Simulations demonstrate 34.6% energy reduction versus offline methods and 2.2% improvement over separated sensing/communication while meeting real-time power grid requirements.

Contribution: First application of ISAC QoS constraints to cooperative UAV trajectory optimization for critical infrastructure monitoring, combining DRL with SDR to handle non-convex ISAC requirements.

Relevance to your work: Tangential - demonstrates ISAC utility in non-traditional use cases and hybrid optimization approaches, but the power grid application and minute-level latency requirements differ substantially from air interface and channel-layer ISAC research.

NTN / satellite

Physics-Informed Digital Twins for Channel Estimation and Traffic Prediction of Non-Terrestrial Networks

Xinyu Huang, Yixiao Zhang, Xue Qin, Mingcheng He, et al. · arXiv eess.SP · 2026-05-22

★★★★■ Addresses two critical NTN challenges with a principled hybrid approach (physics + learning) and demonstrates feasibility on realistic constellation data, though experimental validation remains simulation-based.

This paper develops a physics-informed digital twin framework for LEO satellite networks that addresses channel estimation under sparse/outdated pilots and traffic prediction under heterogeneous demand. The CSI reconstruction uses physics-aware attention with physical-prior tensors to generate full-resolution channel states, while an orbit-adaptive spatiotemporal GNN combines deterministic baseline (from physics) with learned stochastic residuals for traffic forecasting. Validation uses a high-fidelity simulator built on real Starlink ephemeris, global population, and ERA5 weather data.

Contribution: First physics-informed digital twin combining generative CSI reconstruction with spatiotemporal GNN for joint channel-traffic characterization in LEO constellations, validated against realistic orbital and environmental data.

Relevance to your work: Directly relevant: combines your interest in NTN/LEO systems with AI-native channel estimation approaches, and the physics-guided generative modeling resonates with your GAN-based channel work and CSI compression research.